

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MCA (CBCS) Degree Examinations

March 2023

22MCA104 - MATHEMATICAL FOUNDATION FOR COMPUTER APPLICATIONS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from each Unit.*

		Marks	BT*	CO*	PO*
Unit – I					
1. a)	Define a set, an empty set and a singleton set with example for each.	6	L1	1	1
b)	How many positive integers not exceeding 1000 are divisible by 7 or 11?	6	L2	1	1
c)	Find the Eigen values and Eigen vectors of the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$	8	L3	1	2
2. a)	Define union and intersection of two sets with example.	6	L1	1	1
b)	For any three sets A, B, C prove that i) $A \cup B = \overline{A} \cap \overline{B}$ ii) $\overline{A \cap B} = \overline{A} \cup \overline{B}$	6	L3	1	2
c)	A total of 1232 students have taken a course in Spanish, 879 have taken a course in French, and 114 have taken a course in Russian. Further, 103 have taken courses in both Spanish and French, 23 have taken courses in both Spanish and Russian, and 14 have taken courses in both French and Russian. If 2092 students have taken a course in at least one of Spanish French and Russian, how many students have taken a course in all 3 languages.	8	L3	1	2
Unit – II					
3. a)	Let p and q be propositions "The election is decided" and "The votes have been counted" respectively. Express each of the following compound propositions as an English sentence. i) $p \vee q$ ii) $\sim p \wedge q$ iii) $p \leftrightarrow q$	6	L3	2	2
b)	Write the contra positive, the converse, and the inverse of the conditional statement "If the home team wins, then it is raining".	6	L2	2	2
c)	Write the proposition in symbolic form and find its negation. "If all the triangles are right angled then no triangle is equiangular"	8	L3	2	2
4. a)	Using the laws of logic prove the following: $(p \rightarrow q) \wedge [\neg q \wedge (r \vee \neg q)] \Leftrightarrow (q \vee p)$	6	L3	2	2
b)	Prove that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology using truth table.	6	L3	2	2
c)	Show that the compound proposition $[(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p)]$ is logically equivalent to $[(p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow p)]$.	8	L3	2	2
Unit – III					
5. a)	Discuss briefly on equivalence class and partition.	6	L2	3	1
b)	Let $A = \{1, 2, 3, 4\}$ and R be the relation on A defined by xRy iff x divides y. i) Write R as a set of ordered pairs, ii) Draw the diagram of R, iii) Determine in-degrees and out-degrees for each of the vertices.	6	L3	3	2
c)	Prove that for a fixed integer $m > 1$ the relation "congruent modulo m" is an equivalence relation on the set of all integers Z.	8	L3	3	2
6. a)	Let $A = \{1, 2, 3, 4, 5\}$. Define a relation R on $A \times A$ by $(x_1, y_1)R(x_2, y_2)$ iff $x_1 + y_2 = x_2 + y_1$. Verify R is an equivalence relation.	6	L4	3	2
b)	Give the direct graph of the relation $R = \{(1,1), (1,3), (2,1), (2,3), (2,4), (3,1), (3,2), (4,1)\}$ on the set $\{1, 2, 3, 4\}$ also write in-degrees and out-degrees for each of the vertices.	6	L3	3	2

- c) Let R_1 and R_2 are the relations on the set $A = \{1,2,3\}$ represented by the matrices

$$M_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad M_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Determine i) $R_1 \cup R_2$ ii) $R_1 \cap R_2$ iii) R_1 iv) R_2

8 L3 3 2

Unit – IV

7. a) A random variable X takes the values $-3, -2, -1, 0, 1, 2, 3$ such that $P(X=0) = P(X<0)$ and $P(X=-3) = P(X=-2) = P(X=1) = P(X=1) = P(X=2) = P(X=3)$. Determine the Probability distribution.
- b) If the probability of a bad reaction from a certain injection is 0.001, determine the chance that out of 2000 individuals more than two will get a bad reaction.
- c) The probability that a pen manufactured by a company will be defective is $1/10$. If 12 such pens are manufactured, find the probability that
- Exactly two will be defective
 - At least two will be defective
 - None will be defective.
8. a) X is a continuous RV with probability function
- $$f(x) = \begin{cases} kx^2, & -3 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$
- Evaluate k and find i) $P(1 \leq X \leq 2)$, ii) $P(X \leq 2)$.
- b) The length of telephone conversation in a booth has been an exponential distribution and found on an average to be 5 minutes. Find the probability that a random call made from this booth
- ends in less than 5 minutes
 - takes between 5 and 10 minutes.
- c) The marks obtained by 1000 students in an examination are found to be normally distributed with mean 70 and standard deviation 5. Estimate the number of students whose marks will be
- less than 65, ii) more than 75.

6 L3 4 1

7 L3 4 2

7 L3 4 2

6 L4 4 2

7 L3 4 2

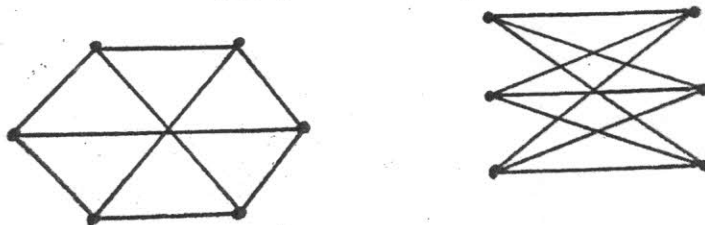
7 L3 4 2

Unit – V

9. a) Define the following with an example.
- Simple graph, ii) Complete graph, iii) Bipartite graph.
- b) Prove that for an undirected graph $G = (V, E)$ the number of vertices of odd degree is even.
- c) Show that the following graphs are isomorphic.

6 L1 5 1

7 L3 5 1



7 L3 5 2

- 10 a) Define the following with an example:
- Circuit, ii) Hamilton path, iii) Euler's path.
- b) Explain Konigsberg Bridge problem and discuss its solution.
- c) Draw the complete bipartite graph: i) $K_{1,3}$ ii) $K_{2,2}$ iii) $K_{3,3}$

6 L1 5 1

7 L2 5 1

7 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
